

Data Dictionary for Driver Identification Downsampled Dataset

File: DriverID_Full_Data_Downsampled.csv

CONTEXT:

The Driver Identification Dataset was made for the purpose of getting various types of data of 50 different drivers. Each driver drove a 2014 Kenworth T270 Class 6 around Fort Collins Collins, Colorado. The data from the CANbus (Control Area Network) was collected during each drive. Additionally, GPS satellite data and inertia data was collected using a VBOX 3i. The route is shown below. A cyberattack is executed on the vehicle during all the drives. This attack caused multiple error lights to turn on in the display and set the tachometer and speedometer to zero, regardless of speed. Drivers were split into three groups. The first group is told nothing about the cyberattack in advance, so it is completely unexpected for them. The second group is given a warning that a cyberattack may or may not occur during the drive. The third group is given a warning that a cyberattack may or may not occur and they are given instructions to slow down and pull over to the side of the road if it occurs. 17 subjects were used in group 1, 16 in group 2, and 17 in group 3. The cyberattack occurs shortly after passing North Overland Trail on Laporte Avenue. The cyberattack is stopped after 1 minute or if the driver pulls over to the side of the road. This data is also used to train machine learning models in order to identify the driver base on how they drive. There is a data variable that shows when in the drive the cyberattack is active. Additionally, there are survey answers from each participant.

This dataset is a downsampled version of the original dataset so that the variables can be grouped to one timestamp. Each variable is downsampled to 1 second by taking the mean of all of the variables' values collected during that second. The standard deviation, min, and max value are calculated and stored to show the variance.

PARTICIPANT IDS WHERE VBOX RESET TOWARDS THE END OF THE DRIVE. ONLY A COUPLE MINUTES OF THE VEHICLE BEING STATIONARY WAS LOST:

- DriverID G1S09

PARTICIPANT IDS WHERE VBOX DATA IS MISSING:

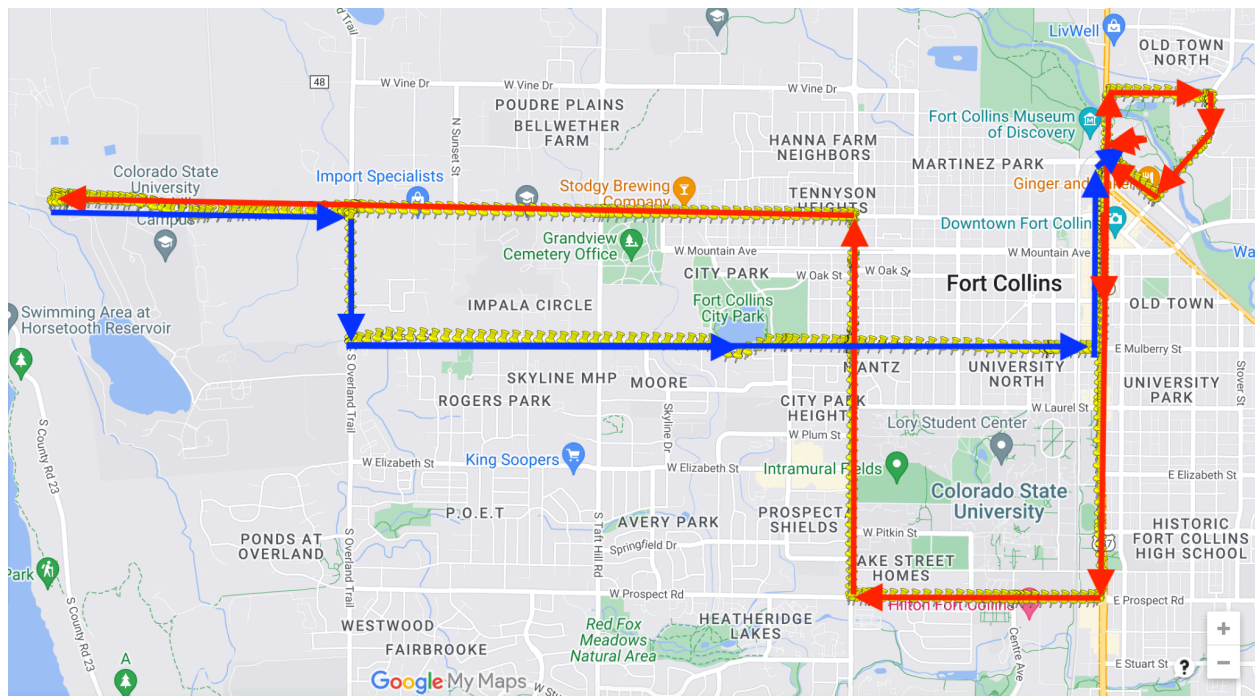
- DriverIDs G1S15,G1S16, G2S13, and G3S13

PARTICIPANT IDS WHERE CANBUS DATA COLLECTED AT LOW FREQUENCY:

- DriverID G1S04

The rest of the participants had successful data collection for both VBOX and CANbus.

Route:



The CANbus data contains CAN (Controller Area Network) data collected from a 2014 Kenworth T270 Class 6 collected from a CANLogger3 and a TruckCape beaglebone black. The data follows the J1939 Standard.

Source:

<https://www.engr.colostate.edu/~jdaily/truck/index.html>
https://www.sae.org/standards/content/j1939da_202301/
<https://github.com/SystemsCyber/TruckCapeProjects>
<https://github.com/SystemsCyber/CAN-Logger-3>

The VBOX Data contains data collected from the VBOX 3i.

Source:

<https://www.vboxautomotive.co.uk/index.php/en/products/data-loggers/vb3i>

The SparkfunGPS Data contains data collected using Teensy 4 and NEO-M9N from SparkFun.

Source:

<https://github.com/SystemsCyber/TruckGPS>

The Heart Rate Monitor Data contains heart rate data collected from an Empatica E4 Wristband

Source:

<https://support.empatica.com/hc/en-us/articles/202581999-E4-wristband-technical-specifications>.

VARIABLE DESCRIPTIONS:

NOTE: All variables from the CANbus, sparkfun GPS, Heart Rate Monitor, and VBOX contain means, standard deviations, mins, and maxes of the total values collected each second. “mean”, “sd”, “max”, and “min” are prepended to all these variables.

Ex:

- **mean.91.Accelerator.Pedal.Position.1** - The mean value of the total values collected during each second.
- **sd.91.Accelerator.Pedal.Position.1** - The standard deviation of the total values collected during each second. If NA is shown, then there was no variance for that second.
- **max.91.Accelerator.Pedal.Position.1** - The max value of the total values collected during each second.
- **min.91.Accelerator.Pedal.Position.1** - The min value of the total values collected during each second.

Data Variables about Participants:

- **ID** - Identification of participant.
 - Ex: G1_Subject1
- **Group** - Group number that participant was placed in for the cyberattack event.
- **Subject** - Subject number in group.

Data Variables about Device Shift and Shift amount. The VBOX did not collect accurate real time values. Due to this, the correlation of the speed collected from the VBOX and the speed collected from the CANbus was calculated. The data from either the VBOX or the CANbus was shifted until it gave the best correlation calculation from their recorded speeds.

- **device_shifted** - Device that was shifted to get the best correlation between speeds from VBOX and CANbus
 - Options:
 - CANbus
 - VBOX
- **shift_amount** - Amount of rows the device from “device_shifted” is moved down.
 - Integer
 - EX: If 10 is the amount and the device_shifted is the VBOX, then that means the VBOX began recording data 10 seconds after the CANbus recorded data. The VBOX data would start on interval 10 and the CANbus starts on interval 0. NA would be shown for all VBOX variables for those 10 seconds.

Data Variables from Heart Rate Monitor:

- **EDA** - Data from the electrodermal activity sensor
 - Units: microsiemens (μS).
- **HR** - Average heart rate extracted from the BVP signal
 - Units: Beats per minute
- **IBI** - Time between individuals' heart beats extracted from the BVP signal.
 - Units: Seconds
 - NOTE: IBI was very inconsistent with its data collection for a good portion of the drives. May not be easy to analyze compared to EDA and HR due to the inconsistency.

Data Variables from the CANbus:

- **Time** - Timestamp corresponding to all data variables from CANbus, sparkfun GPS, Heart Rate Monitor, and VBOX. The format is YEAR-MONTH-DAY HOUR:MINUTE:SECOND. Example: 2024-04-28 11:12:31.
- **Interval_1s** - Seconds interval corresponding to CANbus, sparkfun GPS, Heart Rate Monitor, and VBOX. The drive starts at 0 seconds and the following rows increment by 1 second each until the end of the drive. If the seconds is in the negatives, then that means that it is before the drive began. This portion is where the baseline heart rate of the participant was recorded before the drive began. For this portion, the heart rate will be the only one with values, everything else is NA until 0 seconds when all of the data collection devices were started.
 - Units: Seconds
- **Cyberattack_active** - This signal shows when the cyberattack was activated using a 0 or 1. This was recorded by either using the longitude/latitude values to find the location of when the cyberattack is activated or by using Engine Speed CAN0, which is a separate CAN channel that is directly affected by the cyberattack on the vehicle. When the cyberattack is active, CAN0 goes offline and when it is inactive, CAN0 connects back right away.
 - 0 - Cyberattack not active
 - 1 - Cyberattack active
- **cumulative_distance_meters** - The distance traveled during the drive. This is calculated using the Haversine formula to determine distance on a sphere with latitude and longitude.
 - Units: Meters
- **91.Accelerator.Pedal.Position.1** - This parameter is intended for the primary accelerator control in an application. SPN 91 is the calculated ratio of actual position of the operator's [engine] speed request or [engine] torque request input device (most commonly an accelerator pedal or throttle lever) to the maximum position of the input device. Where redundant means are used to sense the position of the control, SPN 91 shall display the outcome of the manufacturer-defined (i.e. calculated) method to determine the ratio from the redundant inputs.
 - Units: %

- **2945.Active.Shift.Console.Indicator** - Signal from transmission control unit indicating which shift console (primary or secondary) it currently considers as the active shift selector input.
 - 00b = Primary shift console is active
 - 01b = Secondary shift console is active
 - 10b = Reserved
 - 11b = Not available
- **513.Actual Engine...Percent Torque** - The calculated output torque of the engine. The data is transmitted in indicated torque as a percent of reference engine torque (see the engine configuration message, PGN 65251). The engine percent torque value will not be less than zero and it includes the torque developed in the cylinders required to overcome friction.
 - Units: %
- **4154.Actual Engine...Percent.Torque...Fractional.** - This parameter displays an additional torque in percent of the reference engine torque. When the MSB is set to 1, this parameter is not available. When the MSB is set to 0, bits 1 to 3 indicate the desired torque with resolution of 0.125%/bit . Range is from 0 to 0.875% of reference torque. The parameter is used in combination with SPN 513 Actual Engine Torque. The resulting actual engine torque will be calculated by adding these two parameters.
 - 0000b = +0.000%
 - 0001b = +0.125%
 - 0111b = +0.875%
 - 1000b to 1111b = not available
 - For example, if SPN 513 = 150 (or 25%) and this parameter is 0100b, the torque would be 25.5%_x000D_
 - If SPN 513 = 175 (or 50%) and this parameter is 0001b, the torque would be 50.125%_x000D_
 - If SPN 513 = 175 (or 50%) and this parameter is 1111b, the torque would remain 50% (no high resolution available)
- **3357.Actual.Maximum.Available.Engine...Percent.Torque** - This is the maximum amount of torque that the engine can immediately deliver as a percentage of the reference engine torque (SPN 544). The Actual Maximum Available Engine - Percent Torque shall take into consideration all engine torque derates (e.g. air fuel ratio control (AFC), noise control, etc.) that could potentially be active in the system. This parameter differentiates itself from the engine percent torque points 1 through 5 of the engine configuration map because it takes into account all dynamic internal inputs such as AFC and that it is updated on a 50ms basis.
 - Units: %
- **1717.Actual.Maximum.Available.Retarder...Percent.Torque** - This is the maximum amount of torque that the retarder can immediately deliver. It is the same as the maximum torque shown in the Retarder's Configuration message, but allows for a much faster rate of change than could be communicated by reissuing the configuration message.
 - Units: %

- **3031.Aftertreatment.1.Diesel.Exhaust.Fluid.Tank.Temperature.1** - Temperature of the diesel exhaust fluid in the storage tank.
 - Units: °C
- **1761.Aftertreatment.1.Diesel.Exhaust.Fluid.Tank.Volume** - Ratio of volume of diesel exhaust fluid to the total volume of diesel exhaust fluid storage container for aftertreatment system 1 (exhaust bank 1). 0% is empty and 100% is full.
 - Units: %
- **4765.Aftertreatment.1.Diesel.Oxidation.Catalyst.Intake.Temperature** - Temperature of engine combustion byproducts entering the diesel oxidation catalyst in exhaust bank 1. This parameter should be used with engines fueled by diesel fuel (not gaseous fuel like natural gas or propane). A separate parameter has been defined for gas fueled engines.
 - Units: °C
- **3251.Aftertreatment.1.Diesel.Particulate.Filter.Differential.Pressure** - Exhaust differential pressure measured between the intake and exhaust of a diesel particulate filter in exhaust bank 1.
 - Units: kPa
- **3242.Aftertreatment.1.Diesel.Particulate.Filter.Intake.Temperature** - Temperature of engine combustion byproducts entering the diesel particulate filter in exhaust bank 1.
 - Units: °C
- **3610.Aftertreatment.1.Diesel.Particulate.Filter.Outlet.Pressure** - This parameter indicates the diesel particulate filter outlet gage pressure 1 (bank 1).
 - Units: kPa
- **3246.Aftertreatment.1.Diesel.Particulate.Filter.Outlet.Temperature** - Temperature of engine combustion byproducts leaving the diesel particulate filter exhaust in exhaust bank 1.
 - Units: °C
- **5466.Aftertreatment.1.Diesel.Particulate.Filter.Soot.Load.Regeneration.Threshold** - This parameter indicates the value that will first cause DPF regeneration in aftertreatment 1. There may be multiple parameters that cause DPF regeneration (e.g. Percent Soot Load, DPF delta pressure converted to normalized percentage, etc.). If there are multiple inputs to this parameter that may cause active DPF regeneration, the highest value shall always be reported. 100% is the level at which active diesel particulate filter regeneration should be triggered. 100% level is the active regeneration trigger level (and if conditions are not favorable for regeneration, soot loading can continue beyond 100%). During normal operation and regeneration a value 0% will indicate a fully regenerated diesel particulate filter. Values of 25%, 50% and 75% will indicate the general level of soot prior to the 100% level where an active regeneration is needed.
 - Units: %
- **3721.Aftertreatment.1.Diesel.Particulate.Filter.Time.Since.Last.Active.Regeneration** - Indicates the time since the last active regeneration event of diesel particulate filter 1.
 - Units: Seconds

- **3241.Aftertreatment.1.Exhaust.Temperature.1** - The reading from the exhaust temperature sensor located farthest upstream in the aftertreatment system in exhaust bank 1.
 - Units: °C
- **3229.Aftertreatment.1.Outlet.Gas.Sensor.1.at.Temperature** - Indicates that the heater element of the aftertreatment outlet gas sensor, either NO_x or O₂, is within the manufacturer's specified range for accurate measurements in exhaust bank 1.
 - 00b = Not in range
 - 01b = In range
 - 10b = Error
 - 11b = Not available
- **3232.Aftertreatment.1.Outlet.Gas.Sensor.1.Heater.Preliminary.FMI** - Used to identify the applicable J1939-73 FMI detected in the heater of the outlet exhaust gas sensor, either NO_x or O₂, by the manufacturer's sensor control software in exhaust bank 1. In the case of multiple failures the most severe is communicated.
 - A value of 31 is sent to indicate that no failure has been detected or this parameter is not supported.
- **3226.Aftertreatment.1.Outlet.NOx.1** - The amount of combined NO and NO₂ in the exhaust exiting the aftertreatment system measured by a NO_x sensor at the aftertreatment outlet, represented in NO_x molecule parts per million non-NO_x molecules in exhaust bank 1. SPN 3226 shall display the raw, uncorrected NO_x sensor signal. 2016 MY and later engines with NO_x sensors may be required to output a 'corrected' NO_x sensor signal (see SPN 7351) in addition to the raw NO_x sensor signal output using SPN 3226. Any negative values in scaled data from NO_x Sensors may represent positive concentrations that are less than 1 ppm. The negative values shall be treated as 0 ppm (leading to 0 mg NO_x mass) when NO_x mass is to be summed into NO_x Bins provided in PGs 64266, 64267, 64272, 64273, 64278 and 64279.
 - Units: ppm
- **3230.Aftertreatment.1.Outlet.NOx.1.Reading.Stable** - Indicates that the NO_x reading of the aftertreatment outlet NO_x sensor is stable as determined by the manufacturer's control software in exhaust bank 1.
 - 00b = Not stable
 - 01b = Stable
 - 10b = Error
 - 11b = Not available
- **3231.Aftertreatment.1.Outlet.Wide.Range.Percent.Oxygen.1.Reading.Stable** - Indicates that the percent oxygen reading of the aftertreatment outlet gas sensor, either NO_x or O₂, is stable as determined by the manufacturer's control software in exhaust bank 1.
 - 00b = Not stable
 - 01b = Stable
 - 10b = Error
 - 11b = Not available
- **4360.Aftertreatment.1.SCR.Intake.Temperature** - Temperature of engine combustion byproducts entering the SCR in exhaust bank 1.

- Units: °C
- **5862.Aftertreatment.1.SCR.Intermediate.Temperature** - Temperature of engine combustion byproducts at an intermediate point in the SCR component in exhaust bank 1.
 - Units: °C
- **4363.Aftertreatment.1.SCR.Outlet.Temperature** - Temperature of engine combustion byproducts leaving the SCR outlet in exhaust bank 1.
 - Units: °C
- **3239.Aftertreatment.2.Intake.Dew.Point** - Indicates that the temperature on the intake side of the aftertreatment system has exceeded the dew point, as estimated by the ECU in exhaust bank 2.
 - 00b = Not exceeded the dew point
 - 01b = Exceeded the dew point
 - 10b = Error
 - 11b = Not available
- **171.Ambient.Air.Temperature** - Temperature of air surrounding vehicle.
 - Units: °C
- **108.Barometric.Pressure** - Absolute air pressure of the atmosphere.
 - Units: kPa
- **168.Battery.Potential...Power.Input.1** - This parameter measures the first source of battery potential as measured at the input of the ECU/actuator etc. coming from one or more batteries, irrespective of the distance between the component and the battery. This SP is also used when ECUs are interconnected in a series configuration, where the source of power is coming directly or indirectly from the same battery/batteries.
 - Units: V
- **117.Brake.Primary.Pressure** - Gage pressure of air in the primary, or supply side, of the air brake system.
 - Units: kPa
- **118.Brake.Secondary.Pressure** - Gage pressure of air in the secondary, or service side, of the air brake system.
 - Units: kPa
- **5313.Commanded.Engine.Fuel.Rail.Pressure** - Provides the commanded set point value for the engine fuel rail gage pressure
 - Units: MPa
- **7319.Cruise.Control.Speed** - This parameter is the vehicle speed that the cruise control algorithm is currently using as its cruise control set point.
 - Units: km/h
- **512.Driver's.Demand.Engine...Percent.Torque** - The requested torque output of the engine by the driver. It is based on input from the following requestors external to the powertrain: operator (via the accelerator pedal), cruise control and/or road speed limit governor. Dynamic commands from internal powertrain functions such as smoke control, low- and high-speed engine governing; ASR and shift control are excluded from this calculation. The data is transmitted in indicated torque as a percent of the reference engine torque. See PGN 65251 for the engine configuration message. Several status bits

are defined separately to indicate the request which is currently being honored. This parameter may be used for shift scheduling.

- Units: %
- **515.Engine's.Desired.Operating.Speed** - An indication by the engine of the optimal operating speed of the engine for the current existing conditions. These conditions may include the torque generated to accommodate powertrain demands from the operator (via the accelerator pedal), cruise control, road speed limit governors, or ASR. Dynamic commands from functions such as smoke control or shift control are excluded from this calculation.
 - Units: rpm
- **185.Engine.Average.Fuel.Economy** - Average of instantaneous fuel economy for that segment of vehicle operation of interest.
 - Units: km/L
- **110.Engine.Coolant.Temperature** - Temperature of liquid found in engine cooling system.
 - Units: °C
- **101.Engine.Crankcase.Pressure.1** - First instance of the gage pressure inside engine crankcase.
 - Units: kPa
- **2432.Engine.Demand** – Percent Torque - The requested torque output of the engine by all dynamic internal inputs, including smoke control, noise control and low and high speed governing.
 - Units: %
- **1136.Engine.ECU.Temperature** - Temperature of the engine electronic control unit.
 - Units: °C
- **3216.Engine.Exhaust.1.NOx.1** - The amount of combined NO and NO₂ in the exhaust, represented in NO_x molecule parts per million non-NO_x molecules, in exhaust bank 1. This sensor is located at the intake of the SCR system if an SCR system exists.
 - Units: ppm
- **1209.Engine.Exhaust.Bank.1.Pressure.1** - Gage pressure of the exhaust gases as measured at the turbine intake of the turbocharger. This SP to be used for inline engines or exhaust bank 1 of multiple bank engines. See SPN 5749 for exhaust bank 2 and See SPN 6384 for alternate range and resolution.
 - Units: kPa
- **2659.Engine.Exhaust.Gas.Recirculation.1.Mass.Flow.Rate** - Flow rate of gas through the EGR system. Flow rate of the exhaust gas being recirculated into the combustion air.
 - Units: kg/h
- **412.Engine.Exhaust.Gas.Recirculation.1.Temperature** - Temperature of Recirculated Exhaust Gas. This should not be used for Exhaust Gas Recirculation Mixer Intake Temperature. See SPN 5020.
 - Units: °C
- **27.Engine.Exhaust.Gas.Recirculation.1.Valve.Position** - The position of the exhaust gas recirculation valve expressed as a percentage of full travel. Zero percent means the valve

is closed and no exhaust gas is flowing into the intake air stream. One hundred percent means the valve is fully opened.

- Units: %
- **975.Engine.Fan.1.Estimated.Percent.Speed** - Estimated fan speed as a ratio of the fan drive (current speed) to the fully engaged fan drive (maximum fan speed). A two state fan (off/on) will use 0% and 100% respectively. A three state fan (off/intermediate/on) will use 0%, 50% and 100% respectively. A variable speed fan will use 0% to 100%. Multiple fan systems will use 0 to 100% to indicate the percent cooling capacity being provided.
 - Units: %
- **986.Engine.Fan.1.Requested.Percent.Speed** - Fan speed of the first or only fan in the system as a ratio of the actual fan drive (current speed) to the fully engaged fan drive (maximum fan speed). A two state fan (off/on) will use 0% and 100% respectively. A three state fan (off/intermediate/on) will use 0%, 50% and 100% respectively. A variable speed fan will use 0% to 100%. Multiple fan systems will use 0 to 100% to indicate the percent cooling capacity being provided. Feedback to this request is provided using the estimated fan speed (see SPN 975).
 - Units: %
- **157.Engine.Fuel.1.Injector.Metering.Rail.1.Pressure** - The gage pressure of fuel in the primary, or first, metering rail as delivered from the supply pump to the injector metering intake.
 - Units: MPa
- **183.Engine.Fuel Rate** - Amount of fuel consumed by engine per unit of time.
 - Units: l/h
- **184.Engine.Instantaneous.Fuel.Economy** - Current fuel economy at current vehicle velocity.
 - Units: km/L
- **132.Engine.Intake.Air.Mass.Flow.Rate** - Mass flow rate of fresh air entering the engine air intake, before any EGR mixer, if used. Flow rate of fresh air conducted to the engine cylinders to support combustion.
 - Units: kg/h
- **102.Engine.Intake.Manifold..1.Pressure** - The gage pressure measurement of the air intake manifold. If there are multiple air pressure sensors in the intake stream, this is the last one in flow direction before entering the combustion chamber. This should be the pressure used to drive gauges and displays. See also SPNs 1127-1130 and SPN 3562 for alternate range and resolution. If there is only one pressure measurement of the air intake manifold to report and this range and resolution is adequate, this parameter should be used.
 - Units: kPa
- **105.Engine.Intake.Manifold.1.Temperature** - Temperature of pre-combustion air found in intake manifold of engine air supply system.
 - Units: °C
- **100.Engine.Oil.Pressure.1** - Gage pressure of oil in engine lubrication system as provided by the oil pump. See SPN 7468 for alternate resolution.

- Units: kPa
- **92.Engine.Percent.Load.At.Current.Speed** - The ratio of actual engine percent torque (indicated) to maximum indicated torque available at the current engine speed, clipped to zero torque during engine braking.
 - Units: %
- **898.Engine.Requested.Speed.Speed.Limit** - Parameter provided to the engine from external sources in the torque/speed control message. This is the engine speed which the engine is expected to operate at if the speed control mode is active or the engine speed which the engine is not expected to exceed if the speed limit mode is active.
 - Units: rpm
- **4191.Engine.Requested.Torque...Fractional.** - This parameter displays an additional torque in percent of the reference engine torque.
 - 0000b = +0.000%
 - 0001b = +0.125%
 - 0111b = +0.875%
 - 1000b to 1111b = not available
 - For example, if SPN 518 = 150 (or 25%) and this parameter is 0100b, the torque would be 25.5%.
 - If SPN 518 = 175 (or 50%) and this parameter is 0001b, the torque would be 50.125%
 - If SPN 518 = 175 (or 50%) and this parameter is 1111b, the torque would remain 50% (no high resolution available)
 - Units: %
- **518.Engine.Requested.Torque.Torque.Limit** - Parameter provided to the engine or retarder in the torque/speed control message for controlling or limiting the output torque. Requested torque to the engine is measured in indicated torque as a percentage of reference engine torque (see the engine configuration message, PGN 65251). This is the engine torque at which the engine is expected to operate if the torque control mode is active or the engine torque which the engine is not expected to exceed if the torque limit mode is active.
 - Units: %
- **190.Engine.Speed** - Actual engine speed which is calculated over a minimum crankshaft angle of 720 degrees divided by the number of cylinders.
 - Units: rpm
- **190.Engine.Speed.CAN0** - Actual engine speed which is calculated over a minimum crankshaft angle of 720 degrees divided by the number of cylinders collected from CAN0 channel. This is used to find when the cyberattack is active.
 - Units: rpm
- **51.Engine.Throttle.Valve.1.Position.1** - The position of the valve used to regulate the supply of a fluid, usually air or fuel/air mixture, to an engine. 0% represents no supply and 100% is full supply.
 - Units: %
- **250.Engine.Total.Fuel.Used** - Accumulated amount of fuel used during vehicle operation. See SPN 5054 for alternate resolution.

- Units: l
- **247.Engine.Total.Hours.of.Operation** - Accumulated time of operation of engine.
 - Units: hours
- **236.Engine.Total.Idle.Fuel.Used** - Accumulated amount of fuel used during vehicle operation while under idle conditions.
 - Units: l
- **235.Engine.Total.Idle.Hours** - Accumulated time of operation of the engine while under idle conditions.
 - Units: hours
- **249.Engine.Total.Revolutions** - Accumulated number of revolutions of engine crankshaft during its operation.
 - Units: r
- **182.Engine.Trip.Fuel** - Fuel consumed during all or part of a journey. See SPN 5053 for alternate resolution.
 - Units: l
- **1172.Engine.Turbocharger.1.Compressor.Intake.Temperature** - Temperature of the air entering the compressor side of the turbocharger.
 - Units: °C
- **103.Engine.Turbocharger.1.Speed** - Rotational velocity of rotor in the turbocharger.
 - Units: rpm
- **641.Engine.Variable.Geometry.Turbocharger.Actuator..1** - Actuator that controls the variable geometry turbocharger geometry. The control command to the actuator, normalized to percent, where 0% nominally represents fully closed (smallest turbocharger geometry) and 100% represents fully open (largest geometry turbocharger).
 - Units: %
- **2978.Estimated.Engine.Parasitic.Losses...Percent.Torque** - The calculated torque that indicates the estimated amount of torque loss due to engine parasitics, such as cooling fan, air compressor, air conditioning, etc. It is expressed as a percent of Engine Reference Torque.
 - Units: %
- **5398.Estimated.Pumping...Percent.Torque** - The calculated torque that indicates the estimated amount of torque loss due to the engine air handling system. The value transmitted in this parameter is included in SPN 514 (Nominal Friction - Percent Torque). The data is transmitted as a percent of reference engine torque (see the engine configuration message, PGN 65251).
 - Units: %
- **904.Front.Axle.Speed** - The average speed of the two front wheels.
 - Units: km/h
- **96.Fuel.Level.1** - Ratio of volume of fuel to the total volume of fuel storage container.
 - Units: %
- **1242.Instantaneous.Estimated.Brake.Power** - Estimate of the power developed by the engine.
 - Units: kW

- **4207.Message.Checksum** - The message checksum is used to verify the signal path from the transmitting device to the receiving device. The message checksum is calculated using the first 7 data bytes, the message counter and the bytes of the message identifier.
 - Units: count
- **4206.Message.Counter** - The message counter is used to detect situations where the transmitting ECU malfunction repeats the same frame all the time. The receiver of the information may use the counter parameter to detect this situation. The transmitting device will increase the message counter in every cycle. The message counter will count from 0 to 7 and then wrap.
 - Units: count
- **5015.Momentary.Engine.Maximum.Power.Enable** - Momentarily Requesting highest torque map from the engine control
 - 00b = not requesting maximum power available
 - 01b = momentarily requesting maximum power available
 - 10b = fault
 - 11b = not available
- **514.Nominal.Friction...Percent.Torque** - The calculated torque that indicates the amount of torque required by the basic engine itself added by the loss torque of accessories. It contains the frictional and thermodynamic loss of the engine itself, pumping torque loss (SPN 5398), and the losses of fuel, oil and cooling pumps. The data is transmitted as a percent of reference engine torque (see the engine configuration message, PGN 65251).
 - Units: %
- **187.Power.Takeoff.Set.Speed** - Rotational velocity selected by operator for device used to transmit engine power to auxiliary equipment.
 - Units: rpm
- **905.Relative.Speed..Front.Axle..Left.Wheel** - The speed of the front axle, left wheel relative to the front axle speed, SPN 904.
 - Units: km/h
- **906.Relative.Speed..Front.Axle..Right.Wheel** - The speed of the front axle, right wheel relative to the front axle speed, SPN 904.
 - Units: km/h
- **907.Relative.Speed..Rear.Axle..1..Left.Wheel** - The speed of the rear axle #1, left wheel relative to the front axle speed, SPN 904.
 - Units: km/h
- **908.Relative.Speed..Rear.Axle..1..Right.Wheel** - The speed of the rear axle #1, right wheel relative to the front axle speed, SPN 904.
 - Units: km/h
- **1087.Service.Brake.Circuit.1.Air.Pressure** - The pneumatic pressure in the primary service brake circuit or reservoir, supplying the rear axle.
 - Units: kPa
- **1088.Service.Brake.Circuit.2.Air.Pressure** - The pneumatic pressure in the secondary service brake circuit or reservoir, supplying the front axle.
 - Units: kPa

- **3544.Time.Remaining.in.Engine.Operating.State** - This parameter is used to indicate the time remaining in the current engine operating state, based on the state defined in the SP "Engine Operating State" (SPN 3543). For the states in which time remaining is not applicable, use 65535 (FFFFh).
 - Units: Seconds
- **245.Total.Vehicle.Distance** - Accumulated distance traveled by vehicle during its operation.
 - Units: km
- **161.Transmission.1.Input.Shaft.Speed** - Rotational velocity of the primary shaft transferring power into the first transmission. When a torque converter is present, it is the output of the torque converter.
 - Units: rpm
- **191.Transmission.1.Output.Shaft.Speed** - Calculated speed of the transmission 1 output shaft.
 - Units: rpm
- **526.Transmission.Actual.Gear.Ratio** - Actual ratio of input shaft speed to output shaft speed.
 - Units: nan
- **5052.Transmission.Clutch.Converter.Input.Speed** - Rotational velocity of the input to a transmission's master clutch or torque converter. In most cases the rotational velocity will be the same as engine flywheel speed unless there is an intermediate device, such as a power divider, between the engine and the clutch/converter.
 - Units: rpm
- **523.Transmission.Current.Gear** - The gear currently engaged in the transmission or the last gear engaged while the transmission is in the process of shifting to the new or selected gear. Transitions toward a destination gear will not be indicated. Once the selected gear has been engaged then Transmission Current Gear (SPN 523) will reflect that gear.
 - Units: gear value
- **177.Transmission.Oil.Temperature.1** - First instance of transmission lubricant temperature.
 - Units: °C
- **524.Transmission.Selected.Gear** - The gear that the transmission will attempt to achieve during the current shift if a shift is in progress, or the next shift if one is pending (i.e., waiting for torque reduction to initiate the shift).
 - Units: gear value
- **574.Transmission.Shift.In.Process** - Indicates that the transmission is in process of shifting from the current gear to the selected gear. This state is generally ACTIVE during the entire time that the transmission controls the vehicle during a shift. This includes any transmission clutch control, all engine control sequences, pulling to transmission neutral, and engaging the destination gear (e.g., until it is no longer sending commands and/or limits to the engine). See also SPN 560. (See Figure SPN574_A). This state is INACTIVE during other engine control such as torque limiting outside of a shift.
 - 00b = Shift is not in process

- 01b = Shift in process
- 10b = Error
- 11b = Not available
- **573.Transmission.Torque.Converter.Lockup.Engaged** - State signal which indicates whether the torque converter lockup is engaged.
 - 00b = Torque converter lockup disengaged
 - 01b = Torque converter lockup engaged
 - 10b = Error
 - 11b = Not available
- **4816.Transmission.Torque.Converter.Lockup.Transition.in.Process** - State signal indicating whether or not the transmission torque converter lock up clutch is transitioning between being applied and being released. The broadcast value should be set to 01b as soon as an apply or release of the lockup clutch is initiated, and then set to 00b once the given transition is complete. This parameter is a companion to SPN 573 Transmission Torque Converter Lockup Engaged.
 - 00b = Transition is not in process
 - 01b = Transition is in process
 - 10b = Error
 - 11b = Not available
- **3030.Transmission.Torque.Converter.Ratio** - Ratio of the transmissions torque converter output torque to torque converter input torque at current speed.
- **1029.Trip.Average.Fuel.Rate** - Average fuel rate, equal to trip fuel divided by trip time while the engine speed is above zero, since the last trip reset. This includes idle, engine PTO governor (both moving and non-moving) and drive operation but excludes ignition-on time while the engine speed is at zero rpm.
 - Units: l/h
- **244.Trip.Distance** - Distance traveled during all or part of a journey.
 - Units: km
- **918.Trip.Distance..High Resolution.** - Distance traveled during all or part of a journey.
 - Units: m
- **2979.Vehicle.Acceleration.Rate.Limit.Status** - Status (active or not active) of the system used to limit maximum forward vehicle acceleration.
 - 00b = Limit not active
 - 01b = Limit active
 - 10b = Reserved
 - 11b = Not available
- **84.Wheel.Based.Vehicle.Speed** - Speed of the vehicle as calculated from wheel or tailshaft speed.
 - Units: km/h

Data Variables Collected from SparkfunGPSData

- **longitude** - Longitude measured by GPS antenna.
 - Units: Degrees

- **latitude** - Latitude measured by GPS antenna.
 - Units: Degrees
- **gpsSpeed** - Measured velocity from GPS.
 - Units: km/h
- **headingDegree** - Direction of vehicle.
 - Units: Degrees
- **numSats** - Number of satellites detected by GPS antenna.
 - Units: Count of satellites

Data Variables Collected from the RaceLogic VBOX 3i

- **satellites**: The total number of satellites currently being used to calculate position.
 - Units: Count (no units)
- **lat**: The geographic latitude of the VBOX, measured in degrees.
 - Units: Degrees (°)
- **long**: The geographic longitude of the VBOX, measured in degrees.
 - Units: Degrees (°)
- **velocity_kmh**: The speed of the vehicle in kilometers per hour.
 - Units: Kilometers per hour (km/h)
- **heading**: The direction in which the vehicle is traveling, measured in degrees from true north.
 - Units: Degrees (°)
- **height**: The altitude of the VBOX, measured in meters above sea level.
 - Units: Meters (m)
- **vertical_velocity_kmh**: The vertical speed of the vehicle, measured in kilometers per hour.
 - Units: Kilometers per hour (km/h)
- **glonass_sats**: The number of GLONASS (Russian satellite navigation system) satellites being used.
 - Units: Count (no units)
- **gps_sats**: The number of GPS satellites being used.
 - Units: Count (no units)
- **solution type**: The type of GNSS solution being used (e.g., RTK fix, float, 3D, or 2D solution).
 - Units: Descriptive (e.g., RTK, Float)
- **velocity quality kmh**: The quality or confidence level of the velocity measurement, typically based on satellite signal strength.
 - Units: Kilometers per hour (km/h)
- **event 1 time**: The timestamp when a specific event was triggered or recorded.
 - Units: HH:MM (hours, minutes, seconds)
- **Long accel**: Longitudinal acceleration, which measures acceleration along the forward or backward axis of the vehicle.
 - Units: Meters per second squared (m/s²)
- **Lat accel**: Lateral acceleration, which measures acceleration along the side-to-side axis

of the vehicle.

- Units: Meters per second squared (m/s^2)
- **IMU Kalman Filter Status:** The status of the Kalman filter used in the Inertial Measurement Unit (IMU) to improve data accuracy.
 - Units: Descriptive (e.g., Active, Inactive)
- **Head_imu:** The heading of the vehicle, as measured by the IMU.
 - Units: Degrees ($^\circ$)
- **Pitch_imu:** The pitch angle of the vehicle, measured by the IMU.
 - Units: Degrees ($^\circ$)
- **Roll_imu:** The roll angle of the vehicle, measured by the IMU.
 - Units: Degrees ($^\circ$)
- **Pos.Qual.:** Position quality, indicating the accuracy and reliability of the positional data.
 - Units: Descriptive or Relative Value (e.g., Quality Score)
- **Lng_Jerk:** Longitudinal jerk, which measures the rate of change of longitudinal acceleration.
 - Units: Meters per second cubed (m/s^3)
- **Lat_Jerk:** Lateral jerk, which measures the rate of change of lateral acceleration.
 - Units: Meters per second cubed (m/s^3)
- **Head_imu2:** A second heading value from an additional IMU, if present.
 - Units: Degrees ($^\circ$)
- **YawRate:** The rate of rotation around the vertical axis of the vehicle, typically measured in degrees per second.
 - Units: Degrees per second ($^\circ/\text{s}$)
- **X_Accel:** Acceleration along the X-axis of the vehicle (longitudinal direction).
 - Units: Meters per second squared (m/s^2)
- **Y_Accel:** Acceleration along the Y-axis of the vehicle (lateral direction).
 - Units: Meters per second squared (m/s^2)
- **Temp:** Temperature measurement in the vicinity of the VBOX device or its components.
 - Units: Degrees Celsius ($^\circ\text{C}$)
- **PitchRate:** The rate of rotation around the pitch axis, typically measured in degrees per second.
 - Units: Degrees per second ($^\circ/\text{s}$)
- **RollRate:** The rate of rotation around the roll axis, typically measured in degrees per second.
 - Units: Degrees per second ($^\circ/\text{s}$)
- **Z_Accel:** Acceleration along the Z-axis of the vehicle (vertical direction).
 - Units: Meters per second squared (m/s^2)
- **DualStatus:** The status of dual antenna systems, if used, indicating reliability and integrity of the dual GNSS signal.
 - Units: Descriptive (e.g., Active, Inactive)
- **True_Head:** True heading of the vehicle, corrected for magnetic declination.
 - Units: Degrees ($^\circ$)
- **Lat_Vel.:** Lateral velocity of the vehicle, measured in meters per second.
 - Units: Meters per second (m/s)

- **Lng_Vel.:** Longitudinal velocity of the vehicle, measured in meters per second.
 - Units: Meters per second (m/s)
- **RobotHead:** The heading of the vehicle as determined by a robotic control system.
 - Units: Degrees (°)
- **SlipHead:** The slip angle of the vehicle, representing the difference between the heading and actual direction of travel.
 - Units: Degrees (°)
- **Pitch_Ang:** The pitch angle of the vehicle relative to a horizontal plane.
 - Units: Degrees (°)
- **Roll_Angle:** The roll angle of the vehicle relative to a horizontal plane.
 - Units: Degrees (°)
- **Yaw_Rate:** The yaw rate, indicating the rotational velocity around the vertical axis of the vehicle.
 - Units: Degrees per second (°/s)
- **Slip_Angle:** The angle between the vehicle's heading and the trajectory it is following.
 - Units: Degrees (°)
- **T1:** Typically represents an internal timing parameter used for synchronization.
 - Units: Milliseconds (ms)
- **RMS_HPOS:** Root Mean Square error of horizontal position, indicating the precision of horizontal location data.
 - Units: Meters (m)
- **RMS_VPOS:** Root Mean Square error of vertical position, indicating the precision of vertical location data.
 - Units: Meters (m)
- **RMS_HVEL:** Root Mean Square error of horizontal velocity, indicating the precision of horizontal speed data.
 - Units: Meters per second (m/s)
- **RMS_VVEL:** Root Mean Square error of vertical velocity, indicating the precision of vertical speed data.
 - Units: Meters per second (m/s)
- **POSCov_xx, POSCov_xy, POSCov_xz, POSCov_yy, POSCov_yz, POSCov_zz:** Covariance values representing the estimated error between position components.
 - Units: Meters squared (m²)
- **VELCov_xx, VELCov_xy, VELCov_xz, VELCov_yy, VELCov_yz, VELCov_zz:** Covariance values representing the estimated error between velocity components.
 - Units: (Meters/second)² (m²/s²)
- **ms:** Milliseconds portion of the timestamp, providing higher precision time data.
 - Units: Milliseconds (ms)
- **ns:** Nanoseconds portion of the timestamp, providing extremely high precision time data.
 - Units: Nanoseconds (ns)
- **ADC_CH1, ADC_CH2, ADC_CH3, ADC_CH4:** Values from Analog-to-Digital Converter (ADC) channels, measuring external analog inputs.
 - Units: Voltage (V)

- **PreKF Latitude, PreKF Longitude, PreKF Alt, PreKF Velocity, PreKF Heading, PreKF VertSpd:** Pre-Kalman Filter estimates of various parameters, before applying the Kalman filter for smoothing and error correction.
 - Units: Latitude/Longitude: Degrees (°), Altitude: Meters (m), Velocity: Meters per second (m/s), Heading: Degrees (°), Vertical Speed: Meters per second (m/s)
- **Pos_X, Pos_Y, Pos_Z:** Position values in Cartesian coordinates (X, Y, Z).
 - Units: Meters (m)
- **Exported_Elapsed time:** Time elapsed since the start of data recording, typically in seconds.
 - Units: Seconds (s)
- **Exported_Distance:** Total distance covered since the start of data recording.
 - Units: Meters (m)
- **Exported_Longitudinal acceleration:** Longitudinal acceleration calculated after data export.
 - Units: Meters per second squared (m/s^2)
- **Exported_Lateral acceleration:** Lateral acceleration calculated after data export.
 - Units: Meters per second squared (m/s^2)
- **Exported_Relative height:** Relative height change since the start of data recording.
 - Units: Meters (m)
- **Exported_Radius of turn:** Radius of the turn being made by the vehicle, calculated from trajectory data.
 - Units: Meters (m)

Data Variables Relating to Survey from Participant.

NOTE: To save file space, these answers are only stored on second interval 0 from the “Interval_1s” variable. **The answers are not stored on any other second interval.**

- **RecordedDate** - Time and date the survey was filled out.
 - Example format - 4/28/2024 12:05:00 PM
- **Participant.Number** - The participant number is the order of the participants that their drives were done. 1 means they are the first participant that drove the truck. 2 is the second, etc. This does not match the Group or Subject number because some participants in group 3 drove before some in group 2.
- **Experience** - Driving experience of the participant. “Amateur” is for the participants with little to no experience driving a heavy-duty vehicle. “Professional” is for the participants with a commercial drivers license, professional training to drive heavy-duty vehicles, and a high amount of mileage experience driving heavy-duty vehicles.
 - Options: amateur, professional
- **Gender** - Gender of participant
- **Age** - Age of participant
- **Q10: What is the highest level of education you have completed?** - Answer to the question:
 - Options:

- Some high school
 - High school diploma or equivalent
 - Some education beyond high school but no degree
 - College Degree
 - Advanced Degree (e.g., JDS, MS, or PHD)
- **cdl: Have you ever had a Commercial Drivers License (CDL) or Professional Equivalent Certification (i.e. special vehicle training)?** - Answer to the question:
 - Options:
 - Yes, and I still do.
 - Yes, but not currently.
 - No, never.
- **cdl_yrs: How long did you have a CDL or Professional Equivalent Certification (in years)?** - Answer to the question in years.
- **Q20: Do you have experience driving heavy trucks and/or large vehicles?** - Answer to the question:
 - Options
 - Yes
 - No
- **Q21: Briefly describe your experience driving heavy trucks and/or large vehicles.** - Answer to the question if yes is selected for Q20.
 - Answers are typed in by participants.
- **miles: Over the past five years, what is your average annual mileage (estimate all vehicles combined)?** - Answer to question:
 - Options:
 - less than 5,000 miles
 - 5,000 to 10,000 miles
 - 10,000 to 15,000 miles
 - 15,000 to 20,000 miles
 - 20,000 to 25,000 miles
 - more than 25,000 miles
- **tickets: In the past year, how many moving or traffic violations have you had?** - Answer to question:
 - Options
 - 0
 - 1
 - 2
 - 3
 - 4
 - 5
 - more than 5
- **crashes: In the past year, how many crashes have you been in?** - Answer to question:
 - Options:
 - 0
 - 1

- 2
- 3
- 4
- 5
- more than 5

- **driving_yrs: How many years have you been driving?** - Answer to question:
 - Answers are typed in by participants.
- **total_MMDBQ_score** - MMDBQ (Modified Manchester Driver Behavior Questionnaire) total score from Q11. This question is split into 10 parts and an individual score is given based on the answer to each part. The options range from “Never” to “Nearly All the Time”. A score of 1 is given if “Never” is chosen and a score of 6 is given if “Nearly All the Time” is chosen. NA means they did not answer that part. The total score is calculated by adding all of the scores up from all ten parts of the question. The total can range from 10 to 60. The closer the total score is to 60, the more likely the participant may be to take risks while driving. The closer the total score is to 10, the less likely the participant may be to take risks while driving.
- **avg_MMDBQ_score** - MMDBQ (Modified Manchester Driver Behavior Questionnaire) average score from Q11. This question is split into 10 parts and an individual score is given based on the answer to each part. The options range from “Never” to “Nearly All the Time”. A score of 1 is given if “Never” is chosen and a score of 6 is given if “Nearly All the Time” is chosen. NA means they did not answer that part. The average score is calculated by adding all of the scores up from all ten parts of the question and then dividing by **the number of questions that the participant answered**. For example, if the participant only answered 9 of the ten parts, then the average is taken by adding the scores and dividing by 9, **not 10**. The closer the total score is to 6, the more likely the participant may be to take risks while driving. The closer the total score is to 1, the less likely the participant may be to take risks while driving.
- **total_GRiPS_score** - GRiPS (General Risk Propensity Scale) total score from Q12. This question is split into 8 parts and an individual score is given based on the answer to each part. The options range from “Strongly Disagree” to “Strongly Agree”. A score of 1 is given if “Strongly Disagree” is chosen and a score of 5 is given if “Strongly Agree” is chosen. NA means they did not answer that part. The total score is calculated by adding all of the scores up from all eight parts of the question. The total can range from 8 to 40. The closer the total score is to 40, the more likely the participant may be to take risks in general. The closer the total score is to 8, the less likely the participant may be to take risks in general.
- **avg_GRiPS_score** - GRiPS (General Risk Propensity Scale) average score from Q12. This question is split into 8 parts and an individual score is given based on the answer to each part. The options range from “Strongly Disagree” to “Strongly Agree”. A score of 1 is given if “Strongly Disagree” is chosen and a score of 5 is given if “Strongly Agree” is chosen. NA means they did not answer that part. The average score is calculated by adding all of the scores up from all ten parts of the question and then dividing by the **number of questions that the participant answered**. For example, if the participant only answered 7 of the eight parts, then the average is taken by adding the scores and

dividing by 7, **not 8**. The closer the total score is to 5, the more likely the participant may be to take risks while driving. The closer the total score is to 1, the less likely the participant may be to take risks while driving.

- **Q11_1: How often do you do the following while driving... - Attempt to drive away from traffic lights in the wrong gear** - Answer to question:
 - Options:
 - Never
 - Hardly Ever
 - Occasionally
 - Quite Often
 - Frequently
 - Nearly All the Time
- **Q11_2: How often do you do the following while driving... - Become impatient with a slow driver in the fast lane and pass on the right** - Answer to question:
 - Options:
 - Never
 - Hardly Ever
 - Occasionally
 - Quite Often
 - Frequently
 - Nearly All the Time
- **Q11_3: How often do you do the following while driving... - Drive especially close to a car in front as a signal to the driver to go faster or get out of the way** - Answer to question:
 - Options:
 - Never
 - Hardly Ever
 - Occasionally
 - Quite Often
 - Frequently
 - Nearly All the Time
- **Q11_4: How often do you do the following while driving... - Attempt to pass someone that you hadn't noticed to be making a left turn** - Answer to question:
 - Options:
 - Never
 - Hardly Ever
 - Occasionally
 - Quite Often
 - Frequently
 - Nearly All the Time
- **Q11_5: How often do you do the following while driving... - Forget where you left your car in a parking lot** - Answer to question:
 - Options:
 - Never

- Hardly Ever
 - Occasionally
 - Quite Often
 - Frequently
 - Nearly All the Time
- **Q11_6: How often do you do the following while driving... - Turn on one thing, such as your headlights, when you mean to switch on something else, such as the windshield wipers - Answer to question:**
 - Options:
 - Never
 - Hardly Ever
 - Occasionally
 - Quite Often
 - Frequently
 - Nearly All the Time
- **Q11_7: How often do you do the following while driving... - Realize that you have no clear recollection of the road along which you have just been traveling - Answer to question:**
 - Options:
 - Never
 - Hardly Ever
 - Occasionally
 - Quite Often
 - Frequently
 - Nearly All the Time
- **Q11_8: How often do you do the following while driving... - Cross an intersection knowing that the traffic lights have already changed from yellow to red - Answer to question:**
 - Options:
 - Never
 - Hardly Ever
 - Occasionally
 - Quite Often
 - Frequently
 - Nearly All the Time
- **Q11_9: How often do you do the following while driving... - Fail to notice that pedestrians are crossing when turning onto a side street from a main road - Answer to question:**
 - Options:
 - Never
 - Hardly Ever
 - Occasionally
 - Quite Often
 - Frequently

- Nearly All the Time
- **Q11_10: How often do you do the following while driving... - Angered by another driver's behavior, you catch up to them with the intention of giving him/her "a piece of your mind" - Answer to question:**
 - Options:
 - Never
 - Hardly Ever
 - Occasionally
 - Quite Often
 - Frequently
 - Nearly All the Time
- **Q12_1: How strongly do you agree/disagree with the following statements... - Taking risks makes life more fun - Answer to question:**
 - Options:
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **Q12_2: How strongly do you agree/disagree with the following statements... - My friends would say that I'm a risk taker - Answer to question:**
 - Options:
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **Q12_3: How strongly do you agree/disagree with the following statements... - I enjoy taking risks in most aspects of my life - Answer to question:**
 - Options:
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **Q12_4: How strongly do you agree/disagree with the following statements... - I would take a risk even if it meant I might get hurt - Answer to question:**
 - Options:
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **Q12_5: How strongly do you agree/disagree with the following statements... - Taking**

risks is an important part of my life - Answer to question:

- Options:
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **Q12_6: How strongly do you agree/disagree with the following statements... - I commonly make risky decisions** - Answer to question:
 - Options:
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **Q12_7: How strongly do you agree/disagree with the following statements... - I am a believer of taking chances** - Answer to question:
 - Options:
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **Q12_8: How strongly do you agree/disagree with the following statements... - I am attracted, rather than scared, by risk** - Answer to question:
 - Options:
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **cyberattack_1: How strongly do you agree/disagree with the following statements about your study drive by the Foothills campus on Laporte Ave (at the turn around point)... - The vehicle seemed reliable.** - Answer to question:
 - Options:
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **cyberattack_2: How strongly do you agree/disagree with the following statements about your study drive by the Foothills campus on Laporte Ave (at the turn around point)... - The vehicle behaved as expected.** - Answer to question:
 - Options:

- Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **cyberattack_3: How strongly do you agree/disagree with the following statements about your study drive by the Foothills campus on Laporte Ave (at the turn around point)... - I felt safe.** - Answer to question:
 - Options:
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **cyberattack_4: How strongly do you agree/disagree with the following statements about your study drive by the Foothills campus on Laporte Ave (at the turn around point)... - I noticed something wrong with the vehicle.** - Answer to question:
 - Options:
 - Strongly Disagree
 - Somewhat Disagree
 - Neither Agree nor Disagree
 - Somewhat Agree
 - Strongly Agree
- **Q22: Briefly describe what you noticed with the vehicle during the drive.** - Answer to the question.
 - The answer is typed in by the participant.
- **Q11_1_score** - Integer score from part 1 of Q11 based on the participants' answer.
 - Key
 - 1 = "Never"
 - 2 = "Hardly Ever"
 - 3 = "Occasionally"
 - 4 = "Quite Often"
 - 5 = "Frequently"
 - 6 = "Nearly All the Time"
- **Q11_2_score** - Integer score from part 2 of Q11 based on the participants' answer.
 - Key
 - 1 = "Never"
 - 2 = "Hardly Ever"
 - 3 = "Occasionally"
 - 4 = "Quite Often"
 - 5 = "Frequently"
 - 6 = "Nearly All the Time"
- **Q11_3_score** - Integer score from part 3 of Q11 based on the participants' answer.
 - Key

- 1 = "Never"
 - 2 = "Hardly Ever"
 - 3 = "Occasionally"
 - 4 = "Quite Often"
 - 5 = "Frequently"
 - 6 = "Nearly All the Time"
- **Q11_4_score** - Integer score from part 4 of Q11 based on the participants' answer.
 - Key
 - 1 = "Never"
 - 2 = "Hardly Ever"
 - 3 = "Occasionally"
 - 4 = "Quite Often"
 - 5 = "Frequently"
 - 6 = "Nearly All the Time"
- **Q11_5_score** - Integer score from part 5 of Q11 based on the participants' answer.
 - Key
 - 1 = "Never"
 - 2 = "Hardly Ever"
 - 3 = "Occasionally"
 - 4 = "Quite Often"
 - 5 = "Frequently"
 - 6 = "Nearly All the Time"
- **Q11_6_score** - Integer score from part 6 of Q11 based on the participants' answer.
 - Key
 - 1 = "Never"
 - 2 = "Hardly Ever"
 - 3 = "Occasionally"
 - 4 = "Quite Often"
 - 5 = "Frequently"
 - 6 = "Nearly All the Time"
- **Q11_7_score** - Integer score from part 7 of Q11 based on the participants' answer.
 - Key
 - 1 = "Never"
 - 2 = "Hardly Ever"
 - 3 = "Occasionally"
 - 4 = "Quite Often"
 - 5 = "Frequently"
 - 6 = "Nearly All the Time"
- **Q11_8_score** - Integer score from part 8 of Q11 based on the participants' answer.
 - Key
 - 1 = "Never"
 - 2 = "Hardly Ever"
 - 3 = "Occasionally"
 - 4 = "Quite Often"

- 5 = "Frequently"
 - 6 = "Nearly All the Time"
- **Q11_9_score** - Integer score from part 9 of Q11 based on the participants' answer.
 - Key
 - 1 = "Never"
 - 2 = "Hardly Ever"
 - 3 = "Occasionally"
 - 4 = "Quite Often"
 - 5 = "Frequently"
 - 6 = "Nearly All the Time"
- **Q11_10_score** - Integer score from part 10 of Q11 based on the participants' answer.
 - Key
 - 1 = "Never"
 - 2 = "Hardly Ever"
 - 3 = "Occasionally"
 - 4 = "Quite Often"
 - 5 = "Frequently"
 - 6 = "Nearly All the Time"
- **Q12_1_score** - Integer score from part 1 of Q12 based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"
 - 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"
- **Q12_2_score** - Integer score from part 2 of Q12 based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"
 - 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"
- **Q12_3_score** - Integer score from part 3 of Q12 based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"
 - 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"
- **Q12_4_score** - Integer score from part 4 of Q12 based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"

- 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"
- **Q12_5_score** - Integer score from part 5 of Q12 based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"
 - 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"
- **Q12_6_score** - Integer score from part 6 of Q12 based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"
 - 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"
- **Q12_7_score** - Integer score from part 7 of Q12 based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"
 - 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"
- **Q12_8_score** - Integer score from part 8 of Q12 based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"
 - 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"
- **cyberattack_1_score** - Integer score from part 1 of the cyberattack question based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"
 - 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"
- **cyberattack_2_score** - Integer score from part 2 of the cyberattack question based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"

- 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"
- **cyberattack_3_score** - Integer score from part 3 of the cyberattack question based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"
 - 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"
- **cyberattack_4_score** - Integer score from part 4 of the cyberattack question based on the participants' answer.
 - Key
 - 1 = "Strongly Disagree"
 - 2 = "Somewhat Disagree"
 - 3 = "Neither Agree nor Disagree"
 - 4 = "Somewhat Agree"
 - 5 = "Strongly Agree"